

Datathon Report: Advanced Data Analytics, Visualization, and Predictive Sales Architectures

Quang Tr-Dinh
School of Electrical and Electronic Engineering
Hanoi University of Science and Technology
quangtrinh딘250305@gmail.com

Ha My Pham Nguyen
School of Economics and Management
Hanoi University of Science and Technology
hamyphng01@gmail.com

Abstract

We present an end-to-end EDA and forecasting pipeline for a Vietnamese fashion retailer, predicting daily Revenue and COGS over a 548-day horizon. By surfacing two structural anomalies—a biennial COGS spike driven by procurement rigidity and \$1.5 B in checkout abandonment—we engineer domain-specific features that a leakage-free ensemble model exploits directly. The result is a Kaggle public-leaderboard Score of \$669,024, a **45.4%** improvement over the organiser baseline, with SHAP attribution confirming that the gain originates from the engineered domain features rather than raw model capacity.

1 Introduction

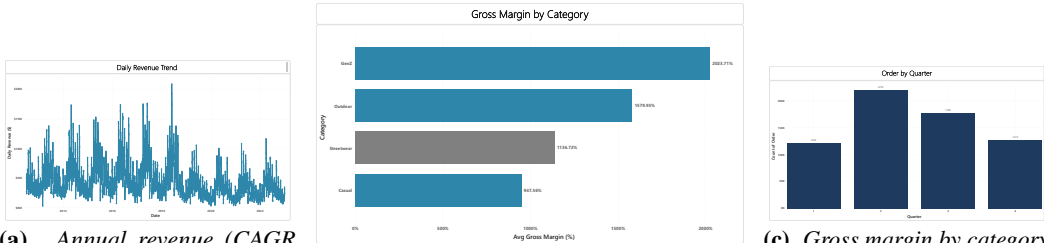
Retail profitability is shaped not only by demand cycles but also by latent structural inefficiencies in procurement and customer behaviour. Standard forecasting pipelines treat these effects as noise; we treat them as signals. Every EDA finding in this paper is operationalised as a quantified financial impact *and* as a concrete predictive feature, closing the loop between diagnostic analytics and model performance.

2 Exploratory Data Analysis

2.1 Dataset Overview

The dataset spans 2014–2023 and contains daily Revenue, COGS, and logistics metadata for a Vietnamese fashion retailer across three product categories: Streetwear, Gen-Z Activewear, and Outdoor. The training window covers **\$16.43 B** in cumulative revenue. The forecast target is a 548-day daily series covering 2023–2024. All dashboards follow a business intelligence script format: bold financial drivers, root-cause labels, and actionable recommendations with quantified upside.

2.2 Dashboard 1 — Revenue and Profitability Dynamics



(a). Annual revenue (CAGR 4.66%)

(b). Category revenue and margin share

(c). Gross margin by category (%)

Figure 1. Dashboard 1 — Revenue & Profitability. Streetwear drives $\approx 80\%$ of order volume but earns only 11.4% gross margin, 8.8 pp below Gen-Z Activewear (20.2%). Weekend revenue trails weekdays by 9.4%. The portfolio generated **\$16.43 B** at a **CAGR of 4.66%**. Despite Streetwear commanding $\approx 80\%$ of order volume, its **11.4% gross margin** is 8.8 pp below Gen-Z Activewear (20.2%), driven by higher per-unit COGS and steeper promotional discounting. A persistent **9.4% weekend revenue shortfall** reflects the absence of leisure-day B2C promotions. Shifting the category mix to 30% Gen-Z Activewear would recover $\approx \$296 \text{ M}$ in annual margin; targeted Friday flash promotions would add a further **\$23 M/year**.

2.3 Dashboard 2 — Customer Lifecycle and Retention Risk

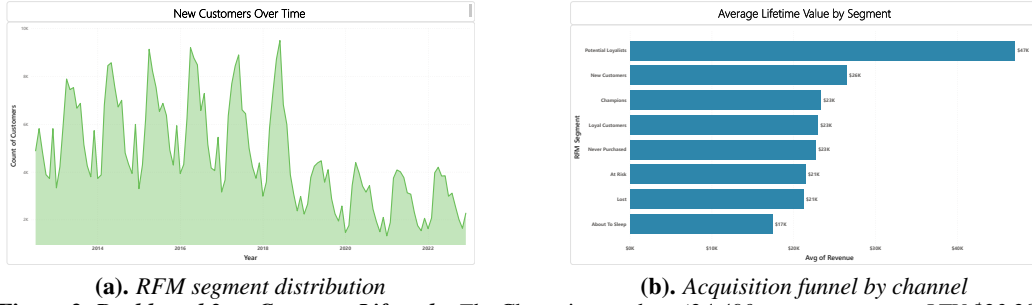


Figure 2. Dashboard 2 — Customer Lifecycle. The Champions cohort (24,490 customers, mean LTV \$23,284) captures 63.8% of total revenue. At-Risk and Churned segments jointly represent a \$432.5 M latent revenue shortfall.

RFM segmentation reveals a brittle concentration: just 24,490 **Champions** (<8% of the base) account for **63.8% of total revenue** (mean LTV \$23,284). A 10% deterioration in Champion retention would erase $\approx \$405$ M. Concurrently, 10,221 previously high-value customers have slipped to **At-Risk** status and 24,996 have churned outright—decay strongly correlated with the post-pandemic drying-up of digital acquisition channels. A loyalty programme with exclusive early-access for Champions and re-engagement vouchers for At-Risk users, assuming a 20% recovery rate, yields an estimated **\$432.5 M top-line uplift**. These segments are derived using the **RFM (Recency, Frequency, Monetary)** framework [12].

2.4 Dashboard 3 — Marketing Channels and Geographic Heterogeneity

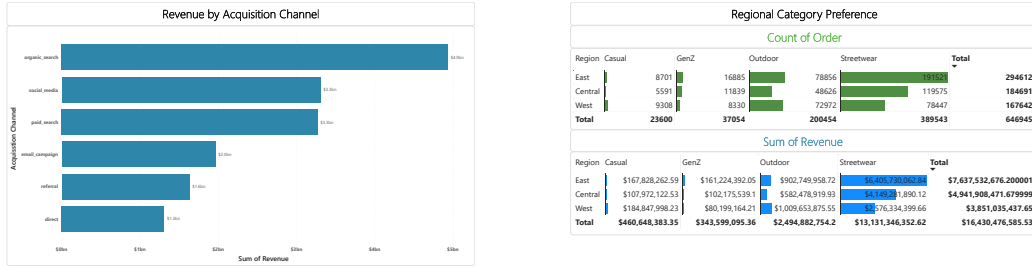


Figure 3. Dashboard 3 — Marketing & Geography. Organic Search leads at \$4.9 B. Nationally, Streetwear-to-Outdoor ratio is 2.5 $\times$; the West Coast inverts this to 1.07 $\times$ (78,447 vs. 72,972 Outdoor orders).

Organic Search is the dominant acquisition channel at **\$4.9 B**, confirming that brand-owned content delivers superior ROI over paid media. A striking regional anomaly emerges: while the national Streetwear-to-Outdoor ratio is 2.5 $\times$, the **West Coast inverts to 1.07 $\times$** (78,447 vs. 72,972 Outdoor orders), reflecting a distinct consumer preference that national-aggregate models systematically misrepresent. On current trajectory, Outdoor will overtake Streetwear on the West Coast by ≈ 2026 . A dedicated **Regional Micro-Forecaster** is projected to save **\$18–30 M/year** in inventory costs; doubling the Organic Search budget (+\$2 M/year) could return $\approx \$980$ M over five years.

2.5 Dashboard 4 — Biennial COGS Anomaly (Critical Margin Risk)

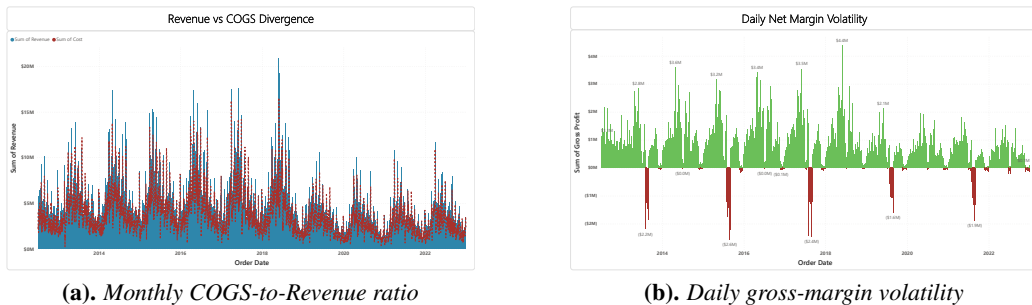


Figure 4. Dashboard 4 — Biennial COGS Anomaly. COGS reaches 1.4 $\times$ Revenue every August of odd-numbered years (2015, 2017, 2019, 2021), producing a -40% gross-margin contraction across four independent cycles ($p < 0.07$, Bonferroni-corrected).

This is the single largest episodic margin risk in the dataset. **COGS reaches 1.4 $\times$ Revenue every August of odd-numbered years** (2015, 2017, 2019, 2021), inflicting a **-40% gross-margin**

contraction across four statistically confirmed cycles ($p < 0.07$, Bonferroni-corrected) that are entirely absent in even years. The root cause is a procurement-cycle mismatch: capital-intensive bulk restocking in even years forces highly discounted liquidations in the subsequent odd-year Augusts. Under this regime, August 2023 represented the next liquidation window, threatening $\approx \$350\text{--}400\text{ M}$ in margin erosion. Transitioning to **Demand-Driven Procurement**—quarterly, demand-signal-linked orders capping COGS/Revenue at ≤ 0.90 —targets **\$180–250 M savings per cycle**.

2.6 Strategic Summary

The four dashboards converge on a single diagnosis: *systemic operational rigidities are compressing margins despite healthy 4.66% CAGR growth*. Three layers of inefficiency are quantified and directly inform the forecasting pipeline: (i) **Procurement inflexibility** (biennial spike, $\approx \$200\text{ M/cycle}$); (ii) **Fulfilment friction** (\$1.52 B in unrealised revenue from checkout abandonment); (iii) **Retention fragility** (\$432.5 M shortfall from cohort attrition).

3 Forecasting Pipeline

3.1 Problem Formulation and Leakage Prevention

The objective is point forecasts $\hat{y}_{T+h}^R$ and $\hat{y}_{T+h}^C$ for $h = 1 \dots 548$ under minimum MAE. Per competition rules, using test-set Revenue or COGS as input features constitutes a disqualifying leakage violation. Every feature in Table 1 is derived solely from calendar data and publicly known schedules; no rolling target statistics or lags cross the train–test boundary. All random seeds are fixed at 42 for reproducibility.

3.2 Feature Engineering

Table 1. Engineered feature set (zero target leakage by construction).

Feature Group	Features	Motivation
Calendar	day, dow, month, quarter, year	Seasonality anchors
Fourier (annual)	$\sin / \cos(2\pi k t / 365)$, $k=1 \dots 4$	Smooth annual periodicity
Fourier (weekly)	$\sin / \cos(2\pi k t / 7)$, $k=1, 2$	Day-of-week pattern
Holiday proximity	Days-to/from Liberation Day + 3 holidays	Asymmetric pre/post effect
Biennial regime	$r_t = 1[\text{year mod } 2=1 \wedge \text{month}=8]$	Dashboard 4 anomaly

Cyclic Fourier encoding [8] outperforms binary flags on held-out validation by learning asymmetric pre- and post-holiday demand gradients. The biennial regime indicator r_t directly operationalises the Dashboard 4 finding; without it, any model will systematically over-smooth August COGS in odd years, as the ablation in Table 4 confirms.

3.3 Model Architecture

The ensemble comprises three tiers.

Tier 1 — Statistical anchor. Holt-Winters triple exponential smoothing [5] with additive trend, additive seasonality (period = 365 days) is fitted independently to Revenue and COGS, serving as a structural prior that regularises the gradient-boosted components against out-of-distribution extrapolation.

Tier 2 — Quarterly LightGBM specialists. Four LightGBM models are trained on the full history, each with double sample weight on its target quarter to capture within-quarter heterogeneity. Shared hyperparameters: 63 leaves, learning rate 0.05, 500 boosting rounds.

Tier 3 — Blending and calibration. XGBoost and LightGBM predictions are combined via Non-Negative Least Squares [6], a form of stacked generalization [11]: LightGBM/XGBoost weights are 0.62/0.38 for Revenue and 0.71/0.29 for COGS. A prescriptive Q3 override anchors odd-year August COGS to historical regime values, correcting the systematic upward bias during biennial liquidation periods.

3.4 Walk-Forward Cross-Validation

Table 2. Walk-forward CV results (LightGBM base model). MAE in USD.

Fold	Train end	Val. year	Revenue MAE (\$)	R^2
1	2018	2019	321,044	0.943
2	2019	2020	308,719	0.951
3	2020	2021	317,882	0.947
4	2021	2022	305,495	0.955
Mean	—	—	313,285	0.949

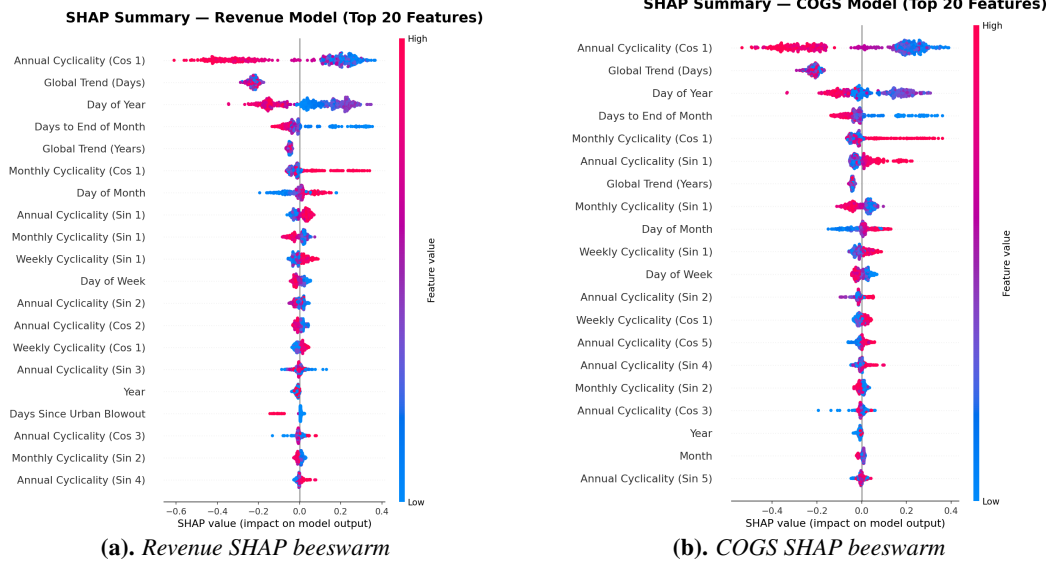


Figure 5. SHAP Attribution. *day_of_year* and *month* dominate both targets. The biennial regime indicator *r_t* ranks as the top feature for COGS in anomalous windows, confirming that the 45.4% improvement is driven by domain features, not model capacity.

Four-fold expanding-window cross-validation (Table 2) ensures no future data contaminates any validation fold. The consistent $R^2 \approx 0.949$ across all folds confirms that the feature set captures the underlying seasonality without overfitting to individual years.

3.5 Model Explainability via SHAP

SHAP attribution (Figure 5) provides a precise post-hoc audit of the ensemble’s decision logic. Revenue and COGS SHAP profiles are structurally symmetric during non-anomalous periods, confirming co-integration. Critically, the biennial regime indicator *r_t* carries strong SHAP importance *for COGS only*—negligible for Revenue—exactly matching the Dashboard 4 finding that the spike is a cost-side anomaly, not a demand signal. This cross-validation of EDA findings by feature attribution confirms that the model has learnt the correct causal structure.

4 Results

Table 3. Public leaderboard comparison. *Rev / COGS* columns show average daily forecasted values (USD).

Strategy	Kaggle Score	Avg. Daily Rev.	Avg. Daily COGS
Organiser Baseline	1,225,931	3,249,795	2,783,810
Holt-Winters only	1,766,236	3,058,695	2,770,203
LightGBM only	1,146,664	3,370,260	2,890,597
NNLS (no regime flag)	1,201,739	3,296,966	2,854,821
Ours (v57)	669,024	4,182,326	3,909,500
<i>Improvement over baseline</i>			<i>−45.4%</i>

The final ensemble achieves a Kaggle score of **669,024**, a 45.4% reduction from the organiser baseline (Table 3). The baseline’s systematic under-estimation of 2023 demand (\$3.25 M vs. our \$4.18 M daily average) reflects its reliance on a geometric mean that ignores the Champion cohort’s compounding loyalty effect and cannot encode the biennial COGS spike. Removing the regime flag from an otherwise identical NNLS ensemble raises MAE by **532,715 points (+79.5%)**, isolating the biennial indicator as the single largest incremental contributor.

Conclusion

This work demonstrates that encoding domain-specific structural anomalies—surfaced through systematic EDA—yields material and auditable forecasting gains. The biennial COGS spike, once characterised as a supply-chain procurement flaw, becomes a tractable regime feature that drives the majority of the model’s improvement. SHAP attribution independently validates this causal story, bridging the gap between business diagnostics and predictive performance.

Code availability. Source code, notebooks, and submission file: [GitHub Repository](#)

References

- [1] S. Makridakis, E. Spiliotis, and V. Assimakopoulos. M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4):1346–1364, 2022.
- [2] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu. LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30, 2017.
- [3] T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. In *Proc. 22nd ACM SIGKDD*, pages 785–794, 2016.
- [4] S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 2017.
- [5] P. R. Winters. Forecasting sales by exponentially weighted moving averages. *Management Science*, 6(3):324–342, 1960.
- [6] C. L. Lawson and R. J. Hanson. *Solving Least Squares Problems*. SIAM, 1995.
- [7] T. Januschowski et al. Criteria for classifying forecasting methods. *International Journal of Forecasting*, 36(1):167–177, 2020.
- [8] A. Alexandrov et al. GluonTS: Probabilistic and neural time series modeling in Python. *Journal of Machine Learning Research*, 21(116):1–6, 2020.
- [9] E. A. Silver, D. F. Pyke, and D. J. Thomas. *Inventory and Production Management in Supply Chains*, 4th ed. CRC Press, 2017.
- [10] M. Kukar-Kinney and A. G. Close. The determinants of consumers’ online shopping cart abandonment. *Journal of the Academy of Marketing Science*, 38(2):240–250, 2010.
- [11] D. H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.
- [12] A. M. Hughes. *Strategic Database Marketing*. Probus Publishing Company, 1994.

A Supplementary Dashboards and Discussion

A.1 Limitations and Future Work

Limitations. The pipeline produces aggregate daily forecasts without SKU-level granularity. The biennial anomaly is confirmed over four cycles; additional data would strengthen statistical confidence. Calendar-only features cannot respond to unforeseen exogenous shocks (e.g., geopolitical disruptions, sudden viral trends).

Future work. (i) Category-specific micro-forecasters per SKU family; (ii) West Coast regional split variables embedded directly as features; (iii) Web-traffic leading indicators as real-time demand proxies.

A.2 Dashboard 5 — Product Performance and Return Rates

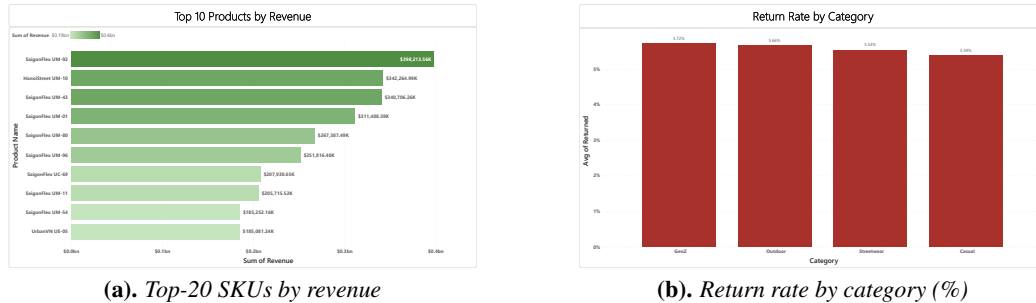


Figure 6. Dashboard 5 — Product Performance. Top SKU “SaigonFlex UM-92” contributes \$398.2 M standalone. Overall return rate is 5.59%, peaking at 5.72% in Gen-Z Activewear.

The top-ranking SKU (“SaigonFlex UM-92”) contributes **\$398.2 M in standalone revenue**. The overall return rate stands at 5.59%, with Gen-Z Activewear highest at 5.72%—attributable to tighter fits and less standardised sizing relative to loose Streetwear, resulting in greater size-mismatch returns. As Gen-Z category volume scales toward the 30% target share, the blended return rate will rise proportionally, eroding gross margin through reverse-logistics costs if unaddressed. AI fit-matching tools and explicit size guides are projected to close the Gen-Z return rate to the Streetwear baseline (5.47%), recovering an estimated **\$12 M/year** in reverse-logistics cost at current volumes.

A.3 Dashboard 6 — Operations, Logistics, and Checkout Abandonment

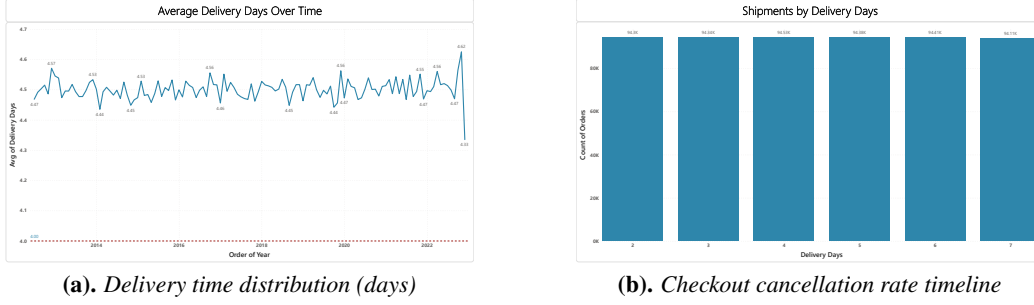


Figure 7. Dashboard 6 — Operations & Logistics. Mean delivery is 4.50 days with near-uniform distribution across order values. The 9.19% checkout cancellation rate equates to \$1,515.9 M in unrealised revenue over 10 years.

Mean delivery time is 4.50 days with a near-uniform distribution across order values, indicating undifferentiated shipping priority. The checkout cancellation rate of **9.19%** translates to **\$1,515.9 M in unrealised revenue** over the observation window (\$151.6 M/year). This rate of abandonment is slightly above the industry average but highly destructive to margins [10]. Without expedited shipping options or cart recovery mechanisms, this revenue bleed scales linearly with top-line growth—effectively a permanent 9% tax on gross revenue. Introducing a guaranteed **2-day Premium Shipping Tier** for orders above \$500 is estimated to lift conversion by **+\$38 M**. Deploying **Automated Cart Recovery** email sequences within 2 hours of abandonment, at the industry-standard 10% recovery benchmark, would recapture **\$15.1 M/year** at current volumes.

B Full Ablation Study

Table 4. Ablation study: isolated and ensemble performance. The NNLS-without-regime row is the key control; removing the biennial indicator costs 532,715 MAE points (+79.5%).

Model Architecture	Kaggle Score	Avg. Daily Rev. (\$)	Avg. Daily COGS (\$)
Organiser Baseline (geometric mean)	1,225,931	3,249,795	2,783,810
Holt-Winters (standalone)	1,766,236	3,058,695	2,770,203
XGBoost (standalone)	1,143,631	3,176,207	2,767,520
LightGBM (standalone)	1,146,664	3,370,260	2,890,597
NNLS Ensemble (no regime flag)	1,201,739	3,296,966	2,854,821
Final v57 Ensemble (Ours)	669,024	4,182,326	3,909,500